

The PICPOT project: GPS and GALILEO in High Dynamic Scenario.

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Abstract— While commercial GPS receivers are able to track objects moving at low velocities (hundreds of kilometres per hour), military applications require special adjustments that allow the tracking system to localize objects moving at high velocities (few kilometres per second) called High Dynamics applications. GALILEO, which represents the European initiative in the field of GNSS (Global Navigation Satellite Systems) is conceived as a civil system. Nonetheless, it is possible imagine that possible improper military applications may be devised. A project is being developed by Politecnico di Torino concerning the study of the characteristics of a GALILEO receiver, compatible with high dynamics (HD) scenarios and especially conceived for working in a space environment, its pros and cons with respect to the GPS signal, in order to develop a cheap system of localization for micro, nano and pico satellites in LEO orbit.

I. INTRODUCTION

The introduction of GPS has represented a revolution in both the military and the civil fields. The primary military purpose was to allow improved command and control of forces through an enhanced ability to accurately specify target locations for troops and to localize nuclear detonation detection by way of special detectors installed in the satellites. Moreover, a large part of modern munitions, the so-called smart bombs or precision-guided munitions, use GPS, but while commercial GPS receivers are able to track objects moving at low velocities (hundreds of kilometres per hour), military applications require special adjustments that allow the tracking system to localize objects moving at high velocities (few kilometres per second) - called High Dynamics applications, due to the high relative motion between the object on which the receiver is installed and the GALILEO Satellites. The most exploited capability of the GPS signal is the P-code pseudo random code which can be used only by U.S. forces, having ten times the frequency of the civilian Coarse/Acquisition (C/A) code and being therefore much more accurate. Nonetheless, the characteristics of the GPS signal, a low-power signals that must travel great distances to reach the receiver, make it extremely vulnerable to both intentional and unintentional interferences (jamming). GPS overcomes many limitations of the smart technologies used in the first Gulf War. While laser-guided bombs, for instance, are utterly lost in smoke or clouds, GPS is far more efficient in those situations.

For all its wonders, though, GPS is far from perfect. The main problem is that the signals transmitted from
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GPS satellites are extremely weak. That was by design, so that the signals wouldn't interfere with other forms of broadcast. But the result is that GPS is easy to jam. A large, powerful GPS jamming station is of limited military use, however, because its transmissions give away its location.

Even without GPS, most or all of the U.S.'s smart weapons can fly because they contain alternative, if less accurate, navigational technologies.

The Galileo system represents the European initiative in the field of GNSS (Global Navigation Satellite Systems), which is expected to be a real boost for the introduction of navigation and positioning applications in the worldwide wireless mass-market. Also many non European countries like China and India and Latin America are expressing a strong interest in participating to Galileo definition, deployment and test. While GPS was originally designed for military applications, GALILEO is conceived as a civil system. Nonetheless, the introduction of the GALILEO localization system and its accessibility for a large number of users introduce the possibility of its possible improper military applications. The team of the Dipartimento di Elettronica at Politecnico di Torino is exploring Galileo characteristics in High Dynamics applications, by way of an experiment of a university pico-satellite (PICPOT PICcolo satellite POLitecnico di Torino) that should be launched in late 2005. Scope of the work is the study of the characteristics of a GALILEO receiver compatible with high dynamics (HD) scenarios and especially conceived for working in a space environment, its pros and cons with respect to the GPS signal, and eventual signal modifications to be suggested to the Galileo Joint Undertaking to avoid improper military use of the signal.

II. GPS SYSTEM

The global positioning system (GPS) is both a military and a civil system, conceived for navigation; it appeared in the 80's and its performances were enhanced by adding more and more satellites (at the moment 28).

Any satellite emits two carriers (L_1, L_2) which are modulated according to a code Division Multiple Access (CDMA) scheme by means of pseudo-random sequences (PN). Each satellite is assigned a different and characterizing PN code. The L_1 carrier (1575.42 MHz) is modulated by the C/A

(coarse acquisition) code, which is a 1023 chip gold code with a chip rate of 1.023 Mbit/s, repeated every 1 ms. A different private code (*P* code) is reserved for military purposes. It makes use of both the carriers, L_1 with a chip rate of 10.23 Mbit/s and L_2 (f 1227.60 MHz), with a chip rate of 10.23 Mbit/s. The *P* code makes use of cryptography techniques and has a length of time of 267 days divided in 37 sections each of seven days. The *C/A* code carries information about the position of the satellite (effemerides) and ionospheric data used by the satellite itself.

There are several sources of error that must be taken into account while computing the user position:

- The geometric dilution of precision (GDOP) involves the uncertainty on the distance from the satellite to the user.
- Until 2000 May 1st, the signal was intentionally merged into noise, in order not to allow the civil users to reach an high accuracy; this is known as the selective “availability” (now disabled).
- The control system involves the uncertainty due to the codes; in this case the use of a longer code, as in the case of the *P* code reserved for military uses, allows an high precision.
- The ionosphere (the distribution of electrons along the path) and the troposphere (temperature, pressure, etc) introduce delays in propagation.
- Uncertainties due to multipath, noise, relativistic effects, and the resolution of the receiver.

III. GALILEO

GPS system gives no guarantee to maintain its quality of service. That's the main reason why an independent European navigation system is needed. Though it's conceived for civil purposes, Galileo presents three different levels of accuracy: the first is for common users; the second offers an enhanced accuracy, under payment; the third is conceived for precision and military applications.

The Galileo system is still in phase of definition by the European community. It has been conceived for civil purposes and it will be operating by 2008. The full constellation will be provided with 27 satellites in medium earth orbit (MEO) and probably 3 geostationary ones (to be confirmed). The GPS system is optimized for the uses at low degrees of latitudes, while, the Galileo system is conceived to be used at every latitude on the earth.

Concerning the transmission scheme, the Galileo system is similar to GPS. Moreover it makes use of the same carrier frequencies (L_1 and L_2). In order to avoid interferences with the GPS signals, a BOC modulation scheme is applied to the GALILEO signals.

IV. GNSS IN HIGH DYNAMICS

Apart from all the aspects related to their suitability to space environment, commercial GPS can hardly work in space environment due to limitations imposed for security issues by the U.S. Government.

The technology of Galileo receivers is still in its initial phase of development, and several projects just started or going to

start in the next months will address the architectural design of the receivers.

Due to the late freezing of the Galileo Signal In Space (SIS) specifications, only few prototypes are going to become available and are usually properties of companies or consortia. Moreover, the possible use of Galileo receivers for high dynamics (e.g. positioning of LEO satellites, sounding rockets, etc...) applications has been so far hardly explored.

From a technological point of view, several issues could potentially arise by the use of the Galileo signals:

- longer codes can have a stronger impact on the acquisition algorithms when a wide range of possible Doppler frequencies has to be considered
- BOC modulation could require modified acquisition algorithms with respect to GPS

In the other hand the richer structure of the GALILEO SIS can be exploited to develop new algorithms and acquisition strategies suitable to high dynamics environments. Scope of this project is the preliminary design and predevelopment of a Galileo SW receiver especially conceived for HD applications, in order to ease Galileo applications in HD environment (space, sounding rockets, ...). Main topics that have to be included in order to adapt GALILEO to space scenario:

- Consolidation of the concepts provided in this proposal by analyzing the market requirements and opportunities, performing a state-of-the-art study on available technology for on board GPS receivers, performing an assessment study of the GALILEO SIS characteristics vs. the required space applications and finally defining the SW receiver testbed configuration.
- Implementation Phase, by customizing the existing simulation environment for being compatible with the SW receiver, performing the analysis, design and implementation of the new algorithms to be integrated into the existing SW receiver prototype (the baseline choice is the GRANADA SW Receiver implemented in the frame of the GARDA project) and finally performing the validation of the receiver.

Notwithstanding the fairly limited time and budget resources, the predevelopment of the GALILEO on-board SW receiver will be possible thanks to the experience of the proposed project team and the availability of facilities like Signal Simulator and SW Receivers developed in the frame of previous GALILEO Joint Undertaking projects, that will be a solid and reliable starting point for the proposed activities. Moreover, the project will address also the definition of the development plan of a technology validation mission dedicated to host on a small satellite a technology demonstrator of the receiver predeveloped in the frame of this project. The goal is to plan such mission in time to exploit the SIS made available by the GALILEO IOV infrastructure.

To scope Galileo advantages and disadvantages with respect to the GPS, an analysis of error sources in space environment has to be done, focusing the attention on the most critical problem in high dynamics scenarios: the Doppler shift, due to the large relative velocities between GPS/Galileo constella-

tions and object to be localized. Galileo characteristics must be explored, in order to find weak points and eventually find a solution for potential problems that can arise with Galileo, and at the same time to concentrate on advantages with respect to GPS to set up an efficient HD Galileo receiver.. As an example, an analysis of acquisition algorithms for Galileo in HD environment must be performed: the longer codes and the BOC modulation are expected to increase the complexity of the acquisition phase. In HD environments this is dramatically reflected in huge delays and processing burden. Those effects must be compensated with efficient algorithms both for the acquisition and for the exploration of the search space (Doppler/delay space). On the other hand this can be traded-off with improved performance, better reliability and robustness. To this aim the features of the Galileo SIS must be investigated (e.g. - Galileo multi frequency receiver should easy the recover of the correct center frequency, not possible in commercial GPS receivers).

It has to be remarked that information regarding commercial implementation of GNSS receivers are always classified, since they are intellectually protected by producers. Therefore, commercial receiver architectures with their implementation choices are not available. However, we can state that the competitiveness on the market from a technological standpoint is given by the ability of performing the receiver procedures with better accuracy, in shorter time and with least complexity as possible. For the purpose of this work, the word standard means that we consider the reference architecture usually proposed in the literature as a baseline scheme.

Galileo (as well as GPS) uses a CDMA (Code Division Multiple Access) technique to broadcast ranging signals from different satellites, employing a Gold code multiplied by a square wave obtaining an equivalent code with a rate equal to 4.092 Mc/s. It is clear that demodulation and tracking of the CDMA signals require a high data processing rate. In addition, a high level of parallel processing is needed since all the code-processing sections have to be replicated per channel. Under these requirements, the high-speed code despreading, the Doppler removal block, and the local code generator have to be implemented on hardware, while for the remaining section a software solution is feasible. In the following paragraphs each block will be shortly explained.

a. **CARRIER AND CODE NUMERICAL CONTROLLED OSCILLATORS** A frequency reference oscillator pilots the SSB modulator of the Doppler removal block. It has to be able to dynamically adapt to the variations given by the Doppler carrier shift. With this aim, a numerical controlled oscillator appears the most effective solution.

b. **CARRIER DISCRIMINATOR AND LOOP FILTER** These two blocks implement a Phase Locked Loop (PLL), which keeps the carrier synchronism tracking the frequency shifts due to the Doppler effect with a resolution of about 1 Hz.

c. **DOPPLER REMOVAL** Owing to satellite clock imperfections and Doppler effects (relative motion between transmitter and receiver), the signal sampled by the A/D converter is affected by a residual modulation. The entity of Doppler

shift is about 50 kHz at a frequency of 1.575 GHz (L1 carrier, used as reference for both GPS and Galileo). This block removes such a spurious modulation through a digital Single Side Band (SSB) modulator, in order to obtain the required phase rotation.

d. **CODE GENERATOR** At the receiver level, the demodulation of incoming CDMA signals broadcasted by Galileo satellites requires a local replica of the spreading codes. With this goal, a code generator able to create a variable delay code according to Galileo systems specification needs to be implemented.

e. **CODE CORRELATOR** This block is in charge of the CDMA demodulation; it removes the spreading effect of the PN codes, via a multiplication by the locally generated and aligned code (which is the output of the code generator).

f. **CODE DISCRIMINATOR AND LOOP FILTER** Once the phase relations have been evaluated, it is necessary to maintain the code alignment through a tracking phase. To this end the Code Discriminator and Loop Filter realize a double phase lock loop (PLL), jointly operating with the Code NCO.

V. THE PICPOT PROJECT

All the problems that arise for the use of GALILEO to locate objects that move at high velocities (High Dynamics) can be evaluated with a proper experiment, like the launch of a low cost satellite.

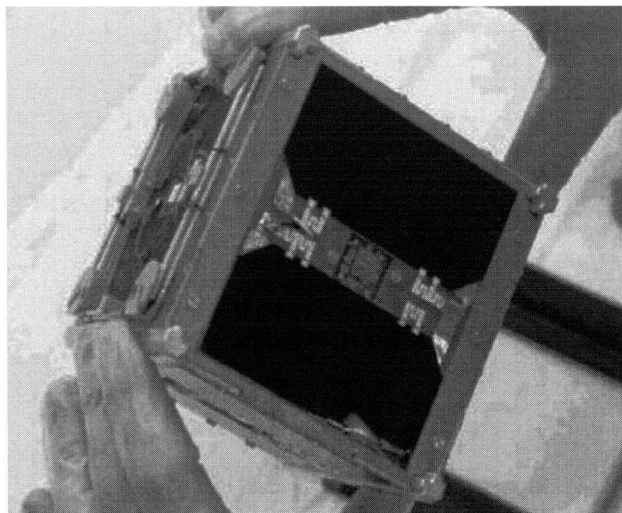


Fig. 1. Image of PicPoT

For accomplishing this activity the following main tasks are being explored:

- define parameters to validate in flight;
- define the validation criteria and strategy: in flight validation campaign;
- derive mission requirements out of in flight validation campaign;
- seek for existing satellite missions that are compatible with such requirements;

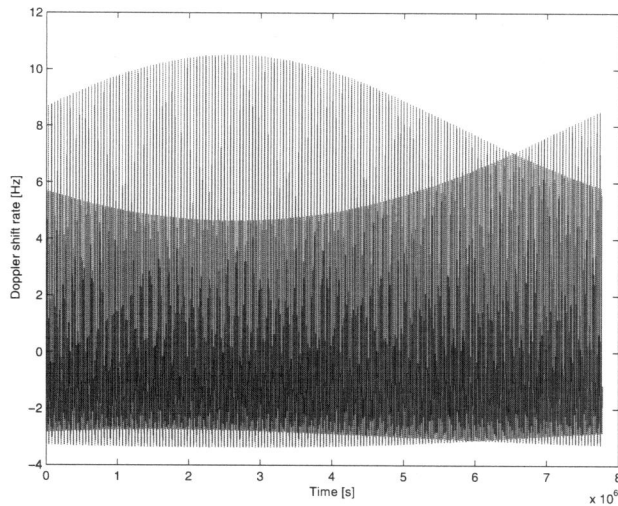


Fig. 2. Doppler shift rate on PicPot, for the GPS signal

- assess and select the most promising satellite mission.

The main parameters to be tested on flight are related to the actual operation of the SW Receiver in space condition and environment. To such purpose, it is necessary to have on board both the SW Receiver and a traditional GPS receiver for comparing and validate the results. Due to the high speed of the satellite, there will be a high doppler shift on the incoming signal.

A simple GPS receiver can't work well; it is necessary to aid the GPS system with a so called doppler-help. This is a circuit which tells the receiver on the satellite how much is the doppler shift on the incoming signal. Simulating previously the real situation, it is possible to store in a memory the values of the quantity of interest. Below there is a plot of the doppler shift rate vs the time, for the supposed orbit of PICPOT.

During the proposed study such parameters and their allowed margins for a successful validation will be listed and justified by the relative rationale.

It is necessary to allow an easy boarding of the in flight validation experiment in order to have as much opportunities as possible of flying it as secondary payload on space missions that are currently being developed. During the proposed study, the durations of the data acquisition slots and the whole validation campaign will be derived, analyzed and optimized according to that.

For the sake of a first iteration, it is possible to anticipate that the driving concept shall have:

- relaxed attitude and orbit requirements;
- short working slots, with no stringent requirement on their timing;
- relaxed downlink of produced data.

Indeed, it is sufficient to have a low earth orbit (LEO) in order to receive the Galileo navigation signals (from the satellites envisaged for the Galileo In Orbit Validation configuration) for most of the orbit and with a wide angle of view with respect to zenith direction.

The short working slots (of the SW Receiver) are mostly aimed at having a very low impact on both the energy budget and the data budget along one orbit. Of course, the larger the capabilities of the hosting platform, the more it is possible to operate the SW Receiver and collect validation data. An applicable mission for boarding the SW Receiver and validating it should therefore have the following features:

- capability to board both the SW Receiver and a GPS receiver (in terms of volume, mass and power)
- having Low Earth Orbit (LEO) characteristics
- have a near zenith pointing attitude for at least one half of the orbit
- having the necessary downlink capability for storing and downloading the acquired data from both SW Receiver and GPS receiver
- having the same timing (in terms of launch date and/or operation duration)

During the proposed study, such features will be further detailed into mission requirements for the SW Receiver validation. Then, following such requirements, it is possible to analyze currently planned missions and assess them for compliance in order to select one of them for carrying the SW Receiver in orbit validation experiment.

Thanks to the relatively small amount of resources needed (a couple of kg and few watts), it is also possible to target mission based on small microsatellites with an overall mass of less than 50 kg.

Such missions are usually of easy access, they are relatively numerous and the required share of the cost is extremely low.

At the moment, according to available SW Receiver data, it is possible to mention two possible opportunities for flying the SW Receiver in orbit validation experiment, and in particular the missions

- PICPOT 2
- Rubin 3

PICPOT 2 is the follow-on of the PICPOT mission (based on the same technology), a university satellite developed by Politecnico di Torino, with cubic shape and 10 cm x 10 cm x 10 cm dimensions.

Such satellite is based on a modular concept, so that it is possible to easily customize it to board the SW Receiver in orbit validation experiment. PICPOT 2 shall have a Sun synchronous orbit of about 700 km and features a magnetic attitude control to be upgraded by a gyro stabilizing momentum wheel.

Its launch could be foreseen around end 2006, provided that the technology validation mission definition is planned and started soon.

Rubin 3 belongs to the low cost family of small microsatellites exploited by Carlo Gavazzi Space (CGS) for validating and space flying small equipments based on new technologies.

Rubin satellites can be launched on any LEO and have an attitude control that goes from spin stabilized to 3 axis stabilized by momentum bias. At the moment a launch date is to be planned, and it could be accommodated in order to

match the availability of the Receiver technology demonstrator and Galileo in orbit validation satellites schedule.

VI. CONCLUSIONS

Within the possibilities of the development of the GALILEO technology, the improper military use has not been explored yet.

In this work we tried to explain the main differences between GPS and GALILEO that are being studied at Politecnico di Torino, and their behavior in High Dynamics scenarios, including GALILEO's employment in LEO Satellites.

An interesting low cost example of a High Dynamics scenario where GALILEO could be employed is the PICPOT project, that includes the use of a GALILEO receiver to locate an object moving at high velocities.

The problems that arise in the PICPOT project are nearly the same of the localization of an object of any type moving at high velocities.

It is clear that in a terrestrial scenario, the low power signal makes jamming easier, like in the case of GPS.

The major goal of the proposed project is the definition and prototyping of a Galileo SW Receiver for Space Applications implemented by an innovative SW Radio approach.

A further outcome of the project is the conceptual design of a flight qualified HW and basic SW platform capable of hosting the developed SW Receiver.

After the design of the receiver, its validation will be made by way of confronting performances with GPS using an hybrid GPS/GALILEO receiver to be installed on the PICPOT that

will be launched in the next months.

In the study we presented the next steps and showed that in theory the use of a GALILEO receiver in high dynamics, is possible in the space scenario, and the PICPOT project is trying to validate this hypothesis with space tests.

VII. ACKNOWLEDGMENTS

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